SIMATS ENGINEERING

ASSESSMENT TOOL 1

Course: Computer Networks (CSA07) | CO5 – Constraint-Based Problem Solving

Q1. Connectivity Solution for a Remote Rural Branch Office

Understanding the Problem & Constraints

The branch office is located where wired broadband (fiber/DSL/cable) is not available. It must still support cloud applications, video conferencing, and secure communication with headquarters, all under a limited budget while maintaining uninterrupted operations. The key constraints are bandwidth availability, cost, reliability, and future scalability.

Recommended Internet Connectivity

- Primary link: 4G/5G LTE wireless broadband via a business-grade cellular router (e.g., Cisco IR1101 or MikroTik LTE router). This avoids the cost of laying new physical cabling and is readily available even in remote areas.
- Backup link: VSAT (satellite internet) or a second SIM from a different telecom provider, configured for automatic failover, to guarantee uninterrupted operation if the primary cellular link degrades or fails.
- A dual-WAN router/firewall (e.g., pfSense or Cisco RV160) manages load balancing and automatic failover between the two links.

Networking Devices & Security Mechanisms

- A managed Layer 2 switch (8–16 ports) connects the office's workstations, printers, and VoIP/video-conferencing endpoints.
- A site-to-site IPsec VPN (or SD-WAN) is established between the branch router and headquarters to encrypt all traffic and give the branch secure access to internal cloud resources.
- A firewall with Unified Threat Management (UTM) features (intrusion prevention, antivirus, content filtering) protects the branch from external threats.
- WPA2/WPA3-Enterprise Wi-Fi with a RADIUS server (or cloud-based RADIUS) authenticates individual users rather than relying on a shared passphrase.
- QoS (Quality of Service) is configured on the router to prioritize video-conferencing and VoIP traffic over general browsing, ensuring call quality despite limited bandwidth.

Justification

Bandwidth limitations are addressed by prioritizing traffic (QoS) and selecting cellular plans with sufficient throughput for video calls (typically 2–4 Mbps per HD call), rather than paying for costly leased lines. Reliability is achieved through dual-WAN failover, so a single link outage does not stop business operations. Cost stays low because 4G/5G and VSAT require no new infrastructure investment compared to fiber trenching, and the dual-WAN router is a one-time capital cost. Scalability is preserved because the same router/firewall can add more SIM cards, higher-tier data plans, or even a wired connection later if the local infrastructure improves, without redesigning the network.

Q2. Campus-Wide Wireless Network for 3,000+ Students

Understanding the Problem & Constraints

The university needs Wi-Fi coverage across classrooms, laboratories, libraries, and hostels for a large, dense population of simultaneous users. The main constraints are a limited number of deployable access points (APs), the need to minimize interference, secure authentication, and overall cost — all while maintaining good coverage and performance.

Access Point Placement Strategy

- Use high-density enterprise APs (e.g., Cisco Cataltic/Aruba/Ubiquiti UniFi) with capacity for 50–100 concurrent clients each, rather than consumer-grade APs, to make the limited AP count go further.
- Place APs based on user density, not just physical area: one AP per classroom/lab of 40–60 students, and 1 AP per ~15–20 hostel rooms (mounted in corridors to serve multiple rooms).
- In libraries and large open halls, mount APs on ceilings in a grid pattern with overlapping coverage of about 10–15% to allow seamless roaming without excessive channel overlap.
- Use directional/sectorized antennas in large auditoriums to focus coverage and reduce co-channel interference from neighboring APs.

Frequency Band Selection

- Deploy dual-band or tri-band APs supporting both 2.4 GHz and 5 GHz (and 6 GHz where Wi-Fi 6E hardware is affordable).
- Use 5 GHz as the primary band for high-density areas (classrooms, library) because it offers more non-overlapping channels (36, 40, 44, 48, 149, 153...) and shorter range, which reduces co-channel interference between adjacent APs.
- Reserve 2.4 GHz mainly for legacy/IoT devices and areas needing longer range with fewer walls, since it has only 3 non-overlapping channels (1, 6, 11) and is more prone to interference.
- Enable band steering so dual-band capable devices are automatically pushed to the less congested 5 GHz band.

Authentication & Security

- Deploy WPA2/WPA3-Enterprise with 802.1X authentication against a centralized RADIUS server integrated with the university's student database, so each student logs in with unique credentials rather than a shared key.
- Use separate SSIDs/VLANs for students, faculty, and guests, with the guest network rate-limited and isolated from internal academic/administrative systems.
- Enable a captive portal for guest access with time-bound tickets, and apply MAC address filtering/monitoring for lab equipment as an added control layer.

Balancing Coverage, Performance, Security, and Cost

Coverage is achieved efficiently by placing high-capacity APs based on real user density rather than uniform spacing, so the limited AP budget covers the busiest areas first. Performance is preserved by using 5 GHz for dense zones (more channels, less interference) and centralized controller-based management that automatically adjusts channel and power settings (RF optimization) as usage patterns change. Security is ensured through WPA2/WPA3-Enterprise with RADIUS-based individual authentication and network segmentation via VLANs, meeting institutional policy without slowing down the network. Cost is controlled by choosing fewer, higher-capacity APs instead of many cheap ones, and by using a central wireless controller that reduces per-AP configuration/management overhead.

Q3. Secure Network Architecture for a Financial Institution

Understanding the Problem & Constraints

The bank must strengthen cybersecurity for sensitive customer data and online banking while satisfying strict regulatory compliance, without materially degrading network performance or inflating operational costs. The constraints therefore span security, performance, compliance, and budget simultaneously.

Network Segmentation

- Divide the network into security zones using VLANs: a DMZ for public-facing services (online banking web servers), an internal zone for employee workstations, a highly restricted zone for core banking servers/databases, and a separate management VLAN for network administration.
- Apply the principle of least privilege between zones using inter-VLAN Access Control Lists (ACLs), so, for example, general staff VLANs cannot directly reach the core database VLAN.
- Isolate ATM/point-of-sale networks and back-office systems into their own segments to contain the impact of any single compromised device.

Firewall Deployment

- Deploy a Next-Generation Firewall (NGFW, e.g., Palo Alto, Fortinet, or Cisco Firepower) at the network perimeter between the internet and the DMZ, providing deep packet inspection, application-layer filtering, and SSL/TLS inspection.
- Place a second internal firewall between the DMZ and the internal/core zones (a two-tier firewall design), so a breach of the public-facing DMZ does not directly expose the core banking systems.
- Configure strict, explicitly-defined rule sets (default-deny) rather than broad allow rules, to minimize the attack surface while keeping legitimate traffic flowing without added latency.

Authentication Methods

- Enforce Multi-Factor Authentication (MFA) for all employee access to sensitive systems and for customer-facing online banking logins, combining passwords with OTP/biometric/hardware tokens.
- Use Role-Based Access Control (RBAC) so employees only access systems relevant to their job function, reducing insider-threat exposure.
- Integrate with a centralized identity provider (e.g., Active Directory with 802.1X/RADIUS) for consistent authentication policy enforcement across wired and wireless access.

Intrusion Detection/Prevention

- Deploy an Intrusion Detection/Prevention System (IDS/IPS) inline at key segment boundaries to detect and block known attack signatures and anomalous traffic patterns in real time.
- Use a Security Information and Event Management (SIEM) system to aggregate logs from firewalls, IDS/IPS, and servers for correlation, alerting, and compliance reporting (e.g., PCI-DSS, RBI/ISO 27001 requirements).
- Conduct regular vulnerability scanning and penetration testing to validate that controls remain effective as the network evolves.

Justification

Security is achieved through layered defenses — segmentation, dual-tier firewalls, MFA, and IDS/IPS — so that no single control failure exposes customer data. Performance impact is minimized by applying deep inspection only at zone boundaries rather than on every internal link, and by using hardware-accelerated NGFWs capable of handling banking-grade throughput without becoming a bottleneck. Compliance is supported because segmentation, MFA, and centralized logging directly map to common regulatory requirements (data isolation, strong authentication, auditability). Budget is respected by prioritizing investment at the perimeter and core-data boundaries — where risk is highest — rather than deploying premium controls uniformly across the entire network, giving the best security return per rupee spent.

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Networking Devices & Security Mechanisms

- A managed Layer 2 switch (8–16 ports) connects the office's workstations, printers, and VoIP/video-conferencing endpoints.
- A site-to-site IPsec VPN (or SD-WAN) is established between the branch router and headquarters to encrypt all traffic and give the branch secure access to internal cloud resources.
- A firewall with Unified Threat Management (UTM) features (intrusion prevention, antivirus, content filtering) protects the branch from external threats.
- WPA2/WPA3-Enterprise Wi-Fi with a RADIUS server (or cloud-based RADIUS) authenticates individual users rather than relying on a shared passphrase.
- QoS (Quality of Service) is configured on the router to prioritize video-conferencing and VoIP traffic over general browsing, ensuring call quality despite limited bandwidth.

Justification

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Frequency Band Selection

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Authentication & Security

- Deploy WPA2/WPA3-Enterprise with 802.1X authentication against a centralized RADIUS server integrated with the university's student database, so each student logs in with unique credentials rather than a shared key.
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- Enable a captive portal for guest access with time-bound tickets, and apply MAC address filtering/monitoring for lab equipment as an added control layer.

Balancing Coverage, Performance, Security, and Cost

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Security is achieved through layered defenses — segmentation, dual-tier firewalls, MFA, and IDS/IPS — so that no single control failure exposes customer data. Performance impact is minimized by applying deep inspection only at zone boundaries rather than on every internal link, and by using hardware-accelerated NGFWs capable of handling banking-grade throughput without becoming a bottleneck. Compliance is supported because segmentation, MFA, and centralized logging directly map to common regulatory requirements (data isolation, strong authentication, auditability). Budget is respected by prioritizing investment at the perimeter and core-data boundaries — where risk is highest — rather than deploying premium controls uniformly across the entire network, giving the best security return per rupee spent.

Q1. Connectivity Solution for a Remote Rural Branch Office

Understanding the Problem & Constraints

The branch office is located where wired broadband (fiber/DSL/cable) is not available. It must still support cloud applications, video conferencing, and secure communication with headquarters, all under a limited budget while maintaining uninterrupted operations. The key constraints are bandwidth availability, cost, reliability, and future scalability.

Recommended Internet Connectivity

- Primary link: 4G/5G LTE wireless broadband via a business-grade cellular router (e.g., Cisco IR1101 or MikroTik LTE router). This avoids the cost of laying new physical cabling and is readily available even in remote areas.
- Backup link: VSAT (satellite internet) or a second SIM from a different telecom provider, configured for automatic failover, to guarantee uninterrupted operation if the primary cellular link degrades or fails.
- A dual-WAN router/firewall (e.g., pfSense or Cisco RV160) manages load balancing and automatic failover between the two links.

Networking Devices & Security Mechanisms

- A managed Layer 2 switch (8–16 ports) connects the office's workstations, printers, and VoIP/video-conferencing endpoints.
- A site-to-site IPsec VPN (or SD-WAN) is established between the branch router and headquarters to encrypt all traffic and give the branch secure access to internal cloud resources.
- A firewall with Unified Threat Management (UTM) features (intrusion prevention, antivirus, content filtering) protects the branch from external threats.
- WPA2/WPA3-Enterprise Wi-Fi with a RADIUS server (or cloud-based RADIUS) authenticates individual users rather than relying on a shared passphrase.
- QoS (Quality of Service) is configured on the router to prioritize video-conferencing and VoIP traffic over general browsing, ensuring call quality despite limited bandwidth.

Justification

Bandwidth limitations are addressed by prioritizing traffic (QoS) and selecting cellular plans with sufficient throughput for video calls (typically 2–4 Mbps per HD call), rather than paying for costly leased lines. Reliability is achieved through dual-WAN failover, so a single link outage does not stop business operations. Cost stays low because 4G/5G and VSAT require no new infrastructure investment compared to fiber trenching, and the dual-WAN router is a one-time capital cost. Scalability is preserved because the same router/firewall can add more SIM cards, higher-tier data plans, or even a wired connection later if the local infrastructure improves, without redesigning the network.

Q2. Campus-Wide Wireless Network for 3,000+ Students

Understanding the Problem & Constraints

The university needs Wi-Fi coverage across classrooms, laboratories, libraries, and hostels for a large, dense population of simultaneous users. The main constraints are a limited number of deployable access points (APs), the need to minimize interference, secure authentication, and overall cost — all while maintaining good coverage and performance.

Access Point Placement Strategy

- Use high-density enterprise APs (e.g., Cisco Cataltic/Aruba/Ubiquiti UniFi) with capacity for 50–100 concurrent clients each, rather than consumer-grade APs, to make the limited AP count go further.
- Place APs based on user density, not just physical area: one AP per classroom/lab of 40–60 students, and 1 AP per ~15–20 hostel rooms (mounted in corridors to serve multiple rooms).
- In libraries and large open halls, mount APs on ceilings in a grid pattern with overlapping coverage of about 10–15% to allow seamless roaming without excessive channel overlap.

- Use directional/sectorized antennas in large auditoriums to focus coverage and reduce co-channel interference from neighboring APs.

Frequency Band Selection

- Deploy dual-band or tri-band APs supporting both 2.4 GHz and 5 GHz (and 6 GHz where Wi-Fi 6E hardware is affordable).
- Use 5 GHz as the primary band for high-density areas (classrooms, library) because it offers more non-overlapping channels (36, 40, 44, 48, 149, 153...) and shorter range, which reduces co-channel interference between adjacent APs.
- Reserve 2.4 GHz mainly for legacy/IoT devices and areas needing longer range with fewer walls, since it has only 3 non-overlapping channels (1, 6, 11) and is more prone to interference.
- Enable band steering so dual-band capable devices are automatically pushed to the less congested 5 GHz band.

Authentication & Security

- Deploy WPA2/WPA3-Enterprise with 802.1X authentication against a centralized RADIUS server integrated with the university's student database, so each student logs in with unique credentials rather than a shared key.
- Use separate SSIDs/VLANs for students, faculty, and guests, with the guest network rate-limited and isolated from internal academic/administrative systems.
- Enable a captive portal for guest access with time-bound tickets, and apply MAC address filtering/monitoring for lab equipment as an added control layer.

Balancing Coverage, Performance, Security, and Cost

Coverage is achieved efficiently by placing high-capacity APs based on real user density rather than uniform spacing, so the limited AP budget covers the busiest areas first. Performance is preserved by using 5 GHz for dense zones (more channels, less interference) and centralized controller-based management that automatically adjusts channel and power settings (RF optimization) as usage patterns change. Security is ensured through WPA2/WPA3-Enterprise with RADIUS-based individual authentication and network segmentation via VLANs, meeting institutional policy without slowing down the network. Cost is controlled by choosing fewer, higher-capacity APs instead of many cheap ones, and by using a central wireless controller that reduces per-AP configuration/management overhead.

Q3. Secure Network Architecture for a Financial Institution

Understanding the Problem & Constraints

The bank must strengthen cybersecurity for sensitive customer data and online banking while satisfying strict regulatory compliance, without materially degrading network performance or inflating operational costs. The constraints therefore span security, performance, compliance, and budget simultaneously.

Network Segmentation

- Divide the network into security zones using VLANs: a DMZ for public-facing services (online banking web servers), an internal zone for employee workstations, a highly restricted zone for core banking servers/databases, and a separate management VLAN for network administration.
- Apply the principle of least privilege between zones using inter-VLAN Access Control Lists (ACLs), so, for example, general staff VLANs cannot directly reach the core database VLAN.
- Isolate ATM/point-of-sale networks and back-office systems into their own segments to contain the impact of any single compromised device.

Firewall Deployment

- Deploy a Next-Generation Firewall (NGFW, e.g., Palo Alto, Fortinet, or Cisco Firepower) at the network perimeter between the internet and the DMZ, providing deep packet inspection, application-layer filtering, and SSL/TLS inspection.
- Place a second internal firewall between the DMZ and the internal/core zones (a two-tier firewall design), so a breach of the public-facing DMZ does not directly expose the core banking systems.
- Configure strict, explicitly-defined rule sets (default-deny) rather than broad allow rules, to minimize the attack surface while keeping legitimate traffic flowing without added latency.

Authentication Methods

- Enforce Multi-Factor Authentication (MFA) for all employee access to sensitive systems and for customer-facing online banking logins, combining passwords with OTP/biometric/hardware tokens.
- Use Role-Based Access Control (RBAC) so employees only access systems relevant to their job function, reducing insider-threat exposure.
- Integrate with a centralized identity provider (e.g., Active Directory with 802.1X/RADIUS) for consistent authentication policy enforcement across wired and wireless access.

Intrusion Detection/Prevention

- Deploy an Intrusion Detection/Prevention System (IDS/IPS) inline at key segment boundaries to detect and block known attack signatures and anomalous traffic patterns in real time.
- Use a Security Information and Event Management (SIEM) system to aggregate logs from firewalls, IDS/IPS, and servers for correlation, alerting, and compliance reporting (e.g., PCI-DSS, RBI/ISO 27001 requirements).
- Conduct regular vulnerability scanning and penetration testing to validate that controls remain effective as the network evolves.

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